Aapa distribution map of Norway

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This script creates a distribution map of appa mires in Norway, for use in reporting for the Bern convention. The resolution is 10x10 km. The distribution is defined based on certain bioclimatic sones and sections.

```
library(tidyverse)
#library(dbConnect)
library(dbplyr)
library(DBI)
library(RPostgres)
library(stars)
library(sf)
library(terra)
library(tmap)
```

Connecting to the database where the bioclimatic sones and sections dataset is

```
# the password is stored in a pgpass.conf file
con <- DBI::dbConnect(drv = RPostgres::Postgres(), host = "t2lippgsql03", dbname = "ano_modul
```

Bring in the data

```
bc <- dplyr::tbl(con, dbplyr::in_schema("helper_variables", "bioclimatic_regions"))

bc <- bc |>
  mutate(geom_wkb = dbplyr::sql("ST_AsBinary(geom)")) |>
  select(-geom) |>
  collect() |>
  mutate(geom = sf::st_as_sfc(geom_wkb, crs = 25833)) |>
  sf::st_as_sf(sf_column_name = "geom")
```

Define distribution

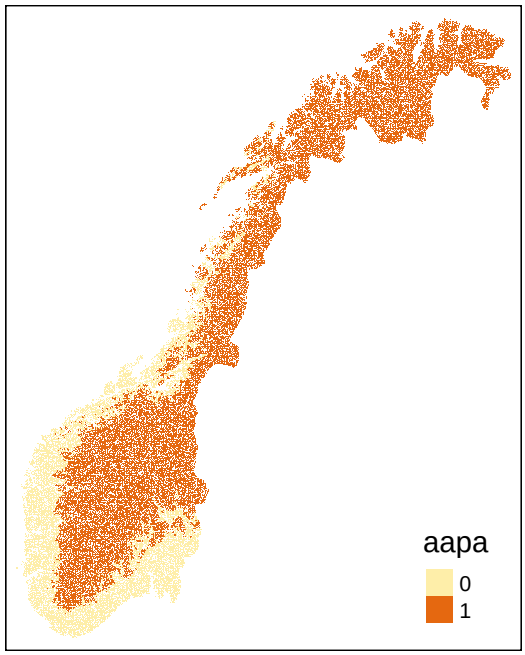
```
reg <- unique(bc$BCregion)
# we exclude boreonemoral and south boreal sones (6SO1-2 and 6SE1)
exl <- grep("6SO-1-2|6SE-1", reg)

# 1 = presence; 0 = absence
bc2 <- bc |>
  mutate(aapa = case_when(
    BCregion %in% reg[exl] ~ 0,
    .default = 1
  ))

# test:
bc2 |> as_tibble() |> count(aapa)
```

```
# A tibble: 2 x 2
   aapa      n
  <dbl>  <int>
1     0  95759
2     1 255643
```

```
# OK
```

A visual test.

```
bc2 |>
  slice_sample(prop = 0.2) |>
  tmap::tm_shape() +
  tmap::tm_fill(col = "aapa", style = "cat")
```



Checking/confirming that the area of the square polygons are 1 km2

```
temp <- bc2 |>
    mutate(km2 = units::drop_units(st_area(geom))*10^-6)

summary(temp |> as_tibble() |> pull(km2))
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1       1       1       1       1       1
```

```
# area is 1 km1
```

```
# defining a summary function
dplyr_summary <- function(data, var) {
  if (inherits(data, "sf")) {
    data <- sf::st_drop_geometry(data)
  }
  data %>%
    summarise(
      Min    = min({{ var }}, na.rm = TRUE),
      Q1     = quantile({{ var }}, 0.25, na.rm = TRUE),
      Median = median({{ var }}, na.rm = TRUE),
```

```
      Mean   = mean({{ var }}, na.rm = TRUE),
      Q3     = quantile({{ var }}, 0.75, na.rm = TRUE),
      Max    = max({{ var }}, na.rm = TRUE)
    )
}
```

Import the EEA 10km standard grid

```
grid <- readRDS(here::here("data", "grid50_kyst_bgr.rds"))

# Test that area is 10 km2
grid |>
  mutate(km2 = units::drop_units(st_area(geometry))*10^-6) |>
  dplyr_summary(km2)
```

```
       Min       Q1  Median     Mean       Q3      Max
1 99.92005 99.98072 100.118 100.1752 100.3278 100.8595
```

```
# the area varies a bit.
```

Creating a regular raster aligned with `grid`.

```
# set resolution
rr <- 10000
# Get bbox
bb <- st_bbox(grid)
st_crs(grid)
```

```
Coordinate Reference System:
  User input: ETRS89 / UTM zone 33N
  wkt:
PROJCRS["ETRS89 / UTM zone 33N",
    BASEGEOGCRS["ETRS89",
        ENSEMBLE["European Terrestrial Reference System 1989 ensemble",
            MEMBER["European Terrestrial Reference Frame 1989"],
            MEMBER["European Terrestrial Reference Frame 1990"],
            MEMBER["European Terrestrial Reference Frame 1991"],
            MEMBER["European Terrestrial Reference Frame 1992"],
            MEMBER["European Terrestrial Reference Frame 1993"],
            MEMBER["European Terrestrial Reference Frame 1994"],
```

```
        MEMBER["European Terrestrial Reference Frame 1996"],
        MEMBER["European Terrestrial Reference Frame 1997"],
        MEMBER["European Terrestrial Reference Frame 2000"],
        MEMBER["European Terrestrial Reference Frame 2005"],
        MEMBER["European Terrestrial Reference Frame 2014"],
        ELLIPSOID["GRS 1980",6378137,298.257222101,
            LENGTHUNIT["metre",1]],
        ENSEMBLEACCURACY[0.1]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    ID["EPSG",4258]],
CONVERSION["UTM zone 33N",
    METHOD["Transverse Mercator",
        ID["EPSG",9807]],
    PARAMETER["Latitude of natural origin",0,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8801]],
    PARAMETER["Longitude of natural origin",15,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8802]],
    PARAMETER["Scale factor at natural origin",0.9996,
        SCALEUNIT["unity",1],
        ID["EPSG",8805]],
    PARAMETER["False easting",500000,
        LENGTHUNIT["metre",1],
        ID["EPSG",8806]],
    PARAMETER["False northing",0,
        LENGTHUNIT["metre",1],
        ID["EPSG",8807]]],
CS[Cartesian,2],
    AXIS["(E)",east,
        ORDER[1],
        LENGTHUNIT["metre",1]],
    AXIS["(N)",north,
        ORDER[2],
        LENGTHUNIT["metre",1]],
USAGE[
    SCOPE["Engineering survey, topographic mapping."],
    AREA["Europe between 12°E and 18°E: Austria; Denmark - offshore and offshore; Germany
    BBOX[46.4,12,84.42,18]],
ID["EPSG",25833]]
```

```r
# create regular grid frm bbox
gr0 <- rast(xmin= floor(bb["xmin"]/rr)*rr, xmax= ceiling(bb["xmax"]/rr)*rr,
              ymin= floor(bb["ymin"]/rr)*rr, ymax= ceiling(bb["ymax"]/rr)*rr,
              res= rr, crs= "EPSG:25833")
ret <- rasterize(grid, gr0, cover=T, touches=T) # (for very small polygons touches=TRUE is ne

class(ret)
```

```
[1] "SpatRaster"
attr(,"package")
[1] "terra"
```

Rasterize and aggregate the 1km2 polygon map

```r
bc_raster_grid <- rasterize(bc2, ret, fun = "max", field = "aapa")
res(bc_raster_grid)
```

```
[1] 10000 10000
```

```r
# OK
```

Import outline of Norway and overlay in the same map to check that the CRS is OK

```r
nor <- readRDS(here::here("data", "norway.rds"))
class(nor)
```

```
[1] "sf"         "tbl_df"     "tbl"        "data.frame"
```

```r
st_crs(nor)
```

```
Coordinate Reference System:
  User input: ETRS89 / UTM zone 33N
  wkt:
PROJCRS["ETRS89 / UTM zone 33N",
    BASEGEOGCRS["ETRS89",
        ENSEMBLE["European Terrestrial Reference System 1989 ensemble",
            MEMBER["European Terrestrial Reference Frame 1989"],
            MEMBER["European Terrestrial Reference Frame 1990"],
            MEMBER["European Terrestrial Reference Frame 1991"],
```

```
        MEMBER["European Terrestrial Reference Frame 1992"],
        MEMBER["European Terrestrial Reference Frame 1993"],
        MEMBER["European Terrestrial Reference Frame 1994"],
        MEMBER["European Terrestrial Reference Frame 1996"],
        MEMBER["European Terrestrial Reference Frame 1997"],
        MEMBER["European Terrestrial Reference Frame 2000"],
        MEMBER["European Terrestrial Reference Frame 2005"],
        MEMBER["European Terrestrial Reference Frame 2014"],
        ELLIPSOID["GRS 1980",6378137,298.257222101,
            LENGTHUNIT["metre",1]],
        ENSEMBLEACCURACY[0.1]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    ID["EPSG",4258]],
CONVERSION["UTM zone 33N",
    METHOD["Transverse Mercator",
        ID["EPSG",9807]],
    PARAMETER["Latitude of natural origin",0,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8801]],
    PARAMETER["Longitude of natural origin",15,
        ANGLEUNIT["degree",0.0174532925199433],
        ID["EPSG",8802]],
    PARAMETER["Scale factor at natural origin",0.9996,
        SCALEUNIT["unity",1],
        ID["EPSG",8805]],
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        LENGTHUNIT["metre",1],
        ID["EPSG",8806]],
    PARAMETER["False northing",0,
        LENGTHUNIT["metre",1],
        ID["EPSG",8807]]],
CS[Cartesian,2],
    AXIS["(E)",east,
        ORDER[1],
        LENGTHUNIT["metre",1]],
    AXIS["(N)",north,
        ORDER[2],
        LENGTHUNIT["metre",1]],
USAGE[
    SCOPE["Engineering survey, topographic mapping."],
    AREA["Europe between 12°E and 18°E: Austria; Denmark - offshore and offshore; German
    BBOX[46.4,12,84.42,18.01]],
```
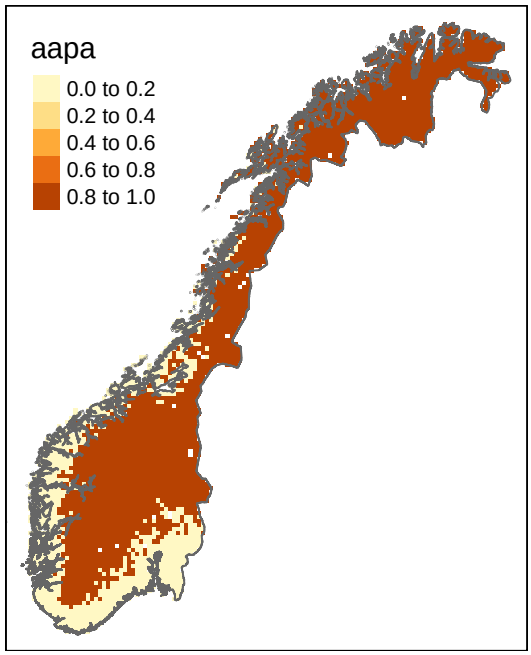

```
    ID["EPSG",25833]]
```

```r
tm_shape(bc_raster_grid)+
  tm_raster("aapa") +
  tm_shape(nor) +
  tm_borders()
```




```r
# OK
```

```r
summary(bc_raster_grid)
```

```
      aapa
 Min.   :0.0000
 1st Qu.:0.0000
 Median :1.0000
 Mean   :0.7239
 3rd Qu.:1.0000
 Max.   :1.0000
 NA's   :14901
```

```r
ncell(bc_raster_grid)
```

```
[1] 18422
```

There are quite a few NA's, but these are from the areas outside mainland Norway (the ocean, Sweden etc.). So that's OK.

I will export in two CRS's.

```r
bc_raster_3035 <- project(bc_raster_grid, "EPSG:3035", method = "near")
```

Exporting the map

```r
writeRaster(bc_raster_grid, "output/aapa_dist_map_25833.tif", overwrite=TRUE)
writeRaster(bc_raster_3035, "output/aapa_dist_map_3035.tif", overwrite=TRUE)
```